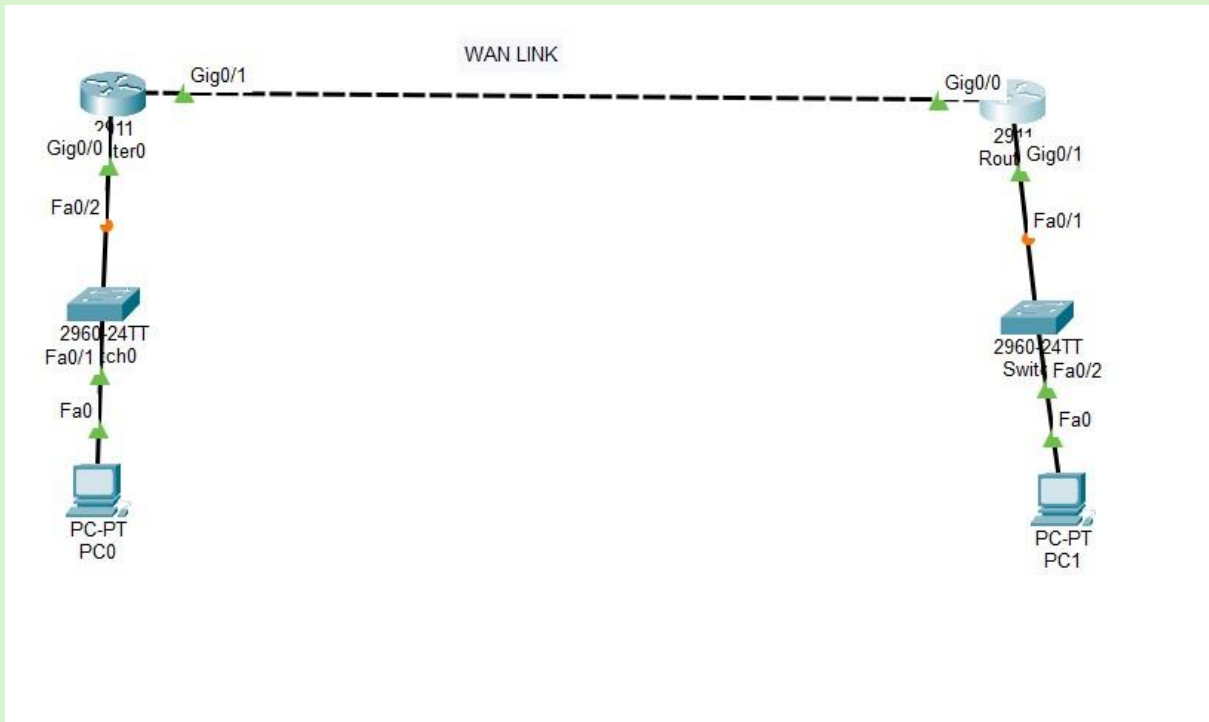


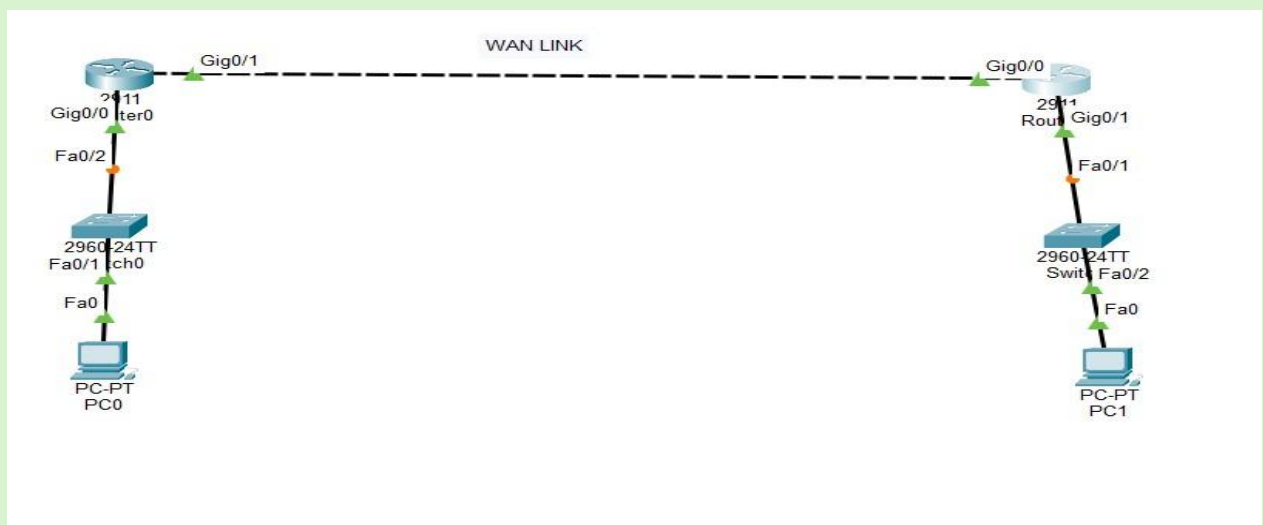
WAN AUTHENTICATION PROTOCOL

1. Draw a simple network diagram showing two routers connected over a WAN link and label where authentication is needed to prevent unauthorized access.



2. Configure a sample HDLC WAN connection between two routers using Packet Tracer or any network simulator, and explain why HDLC does not provide authentication.

Hint: Focus on the commands used and what is missing in terms of security.



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router>ENA
Router#CONF T
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#INTF 50/2/0
Router(config-if)#IP ADD
Router(config-if)#IP Address 10.0.0.1 255.0.0.0
Router(config-if)#en
Router(config-if)#encapsulation hdlc
Router(config-if)#no shut
Router(config-if)#

%LINK-5-CHANGED: Interface Serial0/2/0, changed state to down
Router(config-if)#exit
Router(config)#exit
Router#
$SYS-5-CONFIG I: Configured from console by console

%LINK-5-CHANGED: Interface Serial0/2/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/0, changed state to up

Router#
Router#show ip int br
Router#show ip int brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 unassigned YES unset administratively down down
GigabitEthernet0/1 unassigned YES unset administratively down down
GigabitEthernet0/2 unassigned YES unset administratively down down
Serial0/2/0 10.0.0.1 YES manual up up
Serial0/2/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#show ip int ser
Router#show ip int se
Router#show ip int serial 0/2/0
Serial0/2/0 is up, line protocol is up (connected)
Internet address is 10.0.0.1/8
Broadcast address is 255.255.255.255
Address determined by setup command
MTU is 1500
Helper address is not set
Directed broadcast forwarding is disabled
Outgoing access list is not set
Inbound access list is not set
Proxy ARP is enabled
Security level is default
Split horizon is enabled
ICMP redirects are always sent
ICMP unreachable are always sent
ICMP mask replies are never sent
IP fast switching is disabled
IP fast switching on the same interface is disabled
IP Flow switching is disabled
IP Fast switching turbo vector
```

Why HDLC Does Not Provide Authentication?

HDLC (High-Level Data Link Control) is a Layer 2 WAN encapsulation protocol. It provides framing and data-link communication, but the basic Cisco HDLC protocol does not provide username/password authentication.

There is no command such as:

username Router2 password ... or

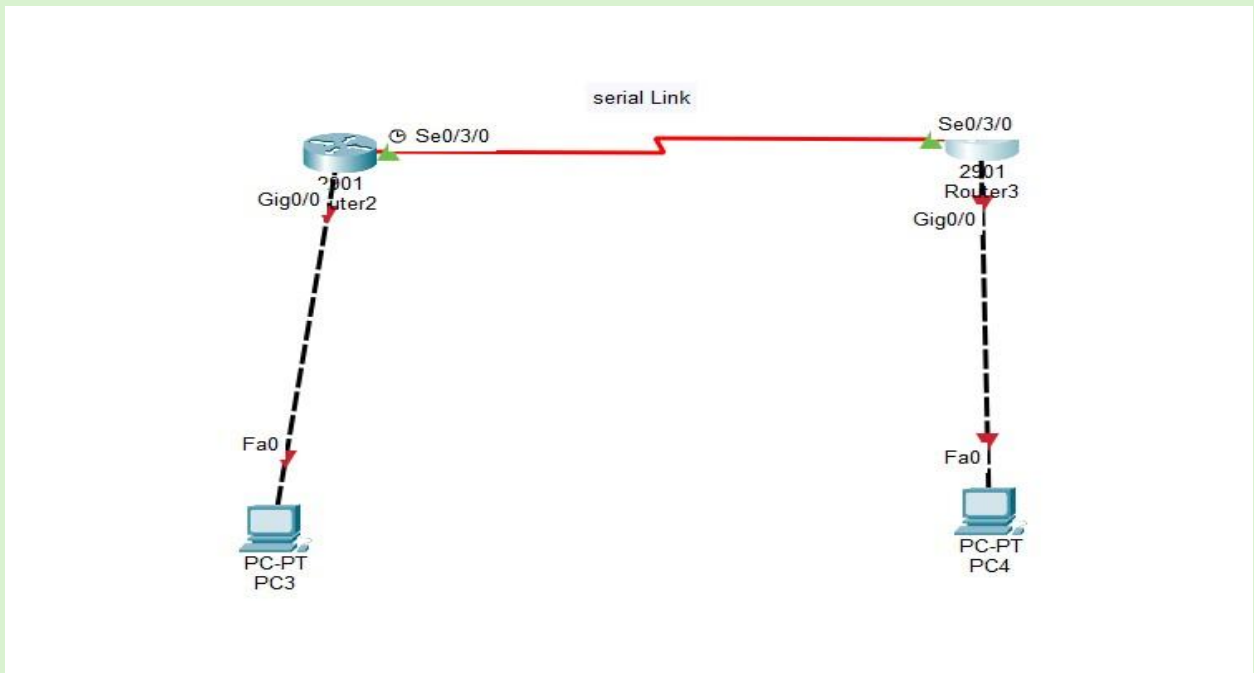
an authentication command like:

ppp authentication chap

in HDLC configuration.

For WAN authentication, PPP can be used instead of HDLC. PPP supports authentication methods such as PAP and CHAP.

3. Set up a PPP connection between two routers in Packet Tracer and enable PAP (Password Authentication Protocol) authentication. Show the configuration commands and explain what happens if the password is incorrect. ○



- R2 Configuration

```
#enable
#configure terminal

#hostname R1

#username R2 password cisco123

#interface serial 0/0/0
#ip address 10.0.0.1 255.255.255.252
#encapsulation ppp
#ppp authentication pap
#ppp pap sent-username R1 password cisco123
#no shutdown
#exit
```

- R3 Configuration

```
#enable
```

```
#configure terminal
```

```
#hostname R2
```

```
#username R1 password cisco123
```

```
#interface serial 0/0/0
```

```
#ip address 10.0.0.2 255.255.255.252
```

```
#encapsulation ppp
```

```
#ppp authentication pap
```

```
#ppp pap sent-username R2 password cisco123
```

```
#no shutdown
```

```
#exit
```

- Verify the Interface On

R1:

```
#show ip interface brief
```

Expected:

```
Serial0/0/0  10.0.0.1  YES manual  up   up
```

On R2:

```
#show ip interface brief
```

Expected:

```
Serial0/0/0  10.0.0.2  YES manual  up   up
```

Both should be:

```
up   up
```

- Test the PPP Connection

From R1:

```
#ping 10.0.0.2
```

If authentication is correct:

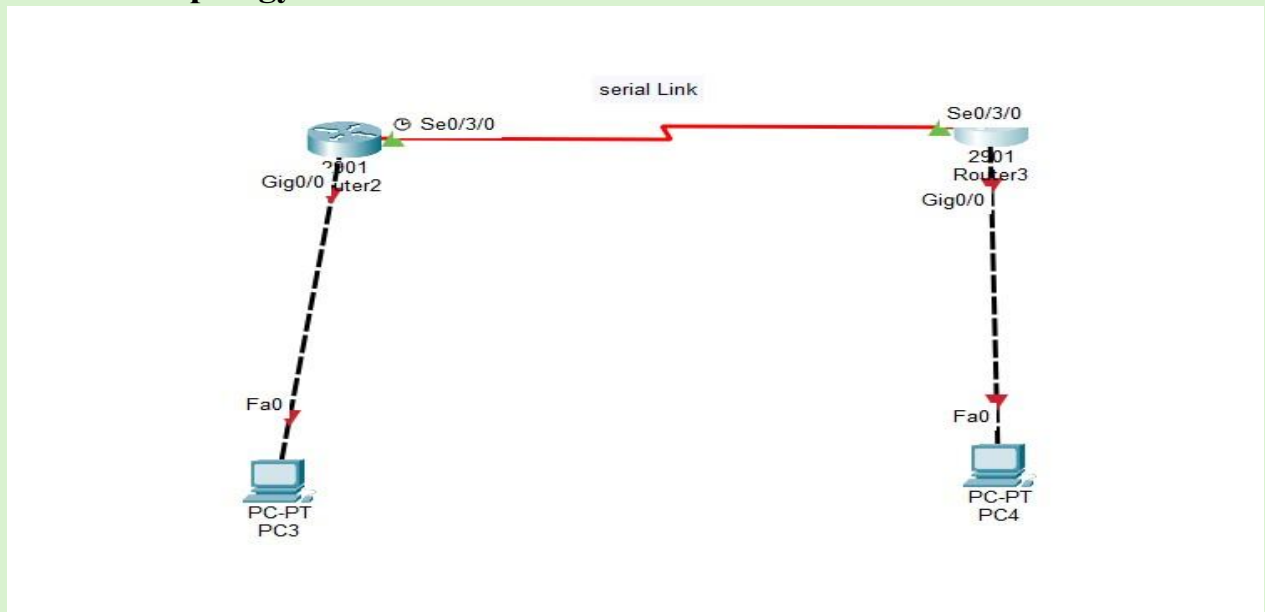
```
!!!!
```

Success rate is 100 percent (5/5)

This means the PPP connection is working successfully.

4. Modify your PPP configuration to use CHAP (Challenge Handshake Authentication Protocol) instead of PAP, and describe one key difference in how CHAP improves security over PAP. ○

**Configure PPP Using CHAP Authentication
Topology**



- **Configuration of R1** enable configure terminal
hostname R1

username R2 password cisco123

interface serial 0/0/0 ip address
10.0.0.1 255.255.255.252
encapsulation ppp ppp
authentication chap no shutdown

exit

- **Configuration of R2** enable configure terminal
hostname R2

username R1 password cisco123

interface serial 0/0/0 ip address
10.0.0.2 255.255.255.252
encapsulation ppp ppp
authentication chap no shutdown

exit

- **Verification**
Check the interface status:
show ip interface brief
- **The Serial interfaces should show:**
up up
- **Check PPP encapsulation:** show interfaces serial
0/0/0

It should display:

Encapsulation PPP

Test the connection:

ping 10.0.0.2

- **Expected output:**
!!!!
Success rate is 100 percent (5/5)

How CHAP Works

CHAP uses a challenge-response authentication process. One router sends a challenge to the other router. The receiving router uses its configured secret/password to generate a response. The first router verifies the response and allows the PPP connection if the credentials are correct.

Difference Between PAP and CHAP

PAP sends the username and password directly for authentication, while CHAP uses a challenge-response mechanism. Therefore, CHAP provides better security because the actual password is not directly transmitted over the PPP link.

- 5. Research and list two real-world scenarios (such as public WiFi login or mobile data authentication) where WAN authentication protocols like PAP, CHAP, or MS-CHAP are used, and explain why authentication is critical in each case.**



Real-World Scenarios of WAN Authentication Protocols

1. Mobile Data Authentication

Mobile network operators use authentication mechanisms such as CHAP or related challenge-response methods to authenticate subscribers and devices before providing network access.

Why authentication is critical:

Authentication ensures that only valid subscribers can access the mobile network. It helps protect user accounts, prevent unauthorized network access, and ensure that network services are provided to the correct customer.

2. Public WiFi Login

Public or enterprise WiFi networks can use authentication protocols such as PAP, CHAP, or MS-CHAP through an authentication server, depending on the network setup. For example, a user may enter a username and password before receiving network access.

Why authentication is critical:

Authentication prevents unauthorized users from accessing the network. It also helps the network administrator identify users, control access, and protect internal network resources from unauthorized connections.